

RESEARCH PAPER

Predictive Modeling for Course Demand and Revenue Forecasting on EduPro

Data-Driven Demand Prediction, Revenue Forecasting, Feature Engineering and Machine Learning Evaluation for an Online Learning Platform

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Project Domain	Educational Technology and Predictive Analytics
Platform	EduPro Online Learning Platform
Primary Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
Models Evaluated	Linear Regression, Ridge, Lasso, Random Forest, Gradient Boosting
Final Enrollment Model	Tuned Random Forest Regressor
Validation	5-Fold Cross Validation
Dataset	Courses, Teachers, Transactions and Users

Research Paper Summary

This paper presents an end-to-end predictive analytics study for EduPro, an online learning platform. The study integrates course, teacher and transaction information to understand enrollment demand and revenue performance. Exploratory analysis is followed by feature engineering, categorical encoding, supervised machine learning, cross-validation and hyperparameter tuning. Particular attention is given to the risk of target leakage, because the initial models produced unrealistically perfect scores. A more defensible enrollment model is subsequently developed using a restricted feature set and a tuned Random Forest Regressor. The final model provides a practical foundation for course planning, marketing prioritization and demand-oriented decision support.

Keywords: predictive analytics; course demand; enrollment forecasting; revenue forecasting; machine learning; Random Forest; online education; feature engineering; cross-validation; EduPro.

1. Abstract and Keywords

Online learning platforms generate large amounts of structured information through course catalogs, instructors, learners and transactions. Converting this information into actionable forecasts can support better course planning, marketing allocation, pricing decisions and resource management. This research develops a predictive modeling framework for EduPro, an online learning platform, with the objective of understanding course demand and revenue behavior and demonstrating how machine learning can support business decision-making.

The study uses four related datasets: 60 courses, 60 teachers, 10,000 transactions and 3,000 users. Initial data-quality assessment identified zero missing values and zero duplicate records in each source table. Course-level enrollment and revenue measures were created from transaction data and merged with the course catalog. Exploratory data analysis examined enrollment distribution, category and level performance, revenue by category, free-versus-paid behavior, correlations, instructor characteristics, transaction trends and user demographics.

Feature engineering created price bands, duration buckets, course rating tiers, instructor experience buckets, teacher rating tiers, expertise-match indicators, revenue per enrollment and a free-course indicator. Categorical variables were encoded for machine learning. Five regression algorithms were evaluated using five-fold cross-validation. Initial results showed extremely high performance from Random Forest and Gradient Boosting. However, these results were investigated because derived variables such as enrollment and revenue-related features can directly encode the target. The research therefore distinguishes between a technically high-scoring model and a methodologically reliable model.

For enrollment prediction, an improved Random Forest model was trained using a smaller set of defensible course and teacher predictors and tuned using GridSearchCV. The selected configuration used 100 trees, maximum depth of 3 and minimum samples per leaf of 2. The tuned model achieved a training R^2 of 0.442 and five-fold cross-validation R^2 of 0.453, with cross-validation MAE of approximately 7.51 enrollments. The gap between training and cross-validation results is substantially more informative than the earlier perfect score and illustrates the importance of leakage-aware modeling.

The research concludes that EduPro can benefit from predictive analytics, but model quality must be evaluated together with feature availability, leakage risk, validation design and business interpretability. The proposed framework can be extended with time-based validation, learner-level behavior, campaign variables and future-period forecasting to produce a more operational demand forecasting system.

Keywords: EduPro, predictive modeling, enrollment demand, course revenue, machine learning, Random Forest, feature engineering, cross-validation, educational analytics, business intelligence.

2. Introduction

2.1 Background

Digital education has transformed the way learners discover, purchase and consume educational content. Online platforms typically maintain a course catalog containing attributes such as category, level, price, duration and rating. They also maintain instructor information and transaction histories. These datasets can be analyzed jointly to identify demand patterns and predict future outcomes. Predictive analytics is particularly useful when an organization must make decisions before outcomes are known.

For an online learning platform, course demand is not determined by a single variable. Price can influence purchase decisions, but so can perceived quality, course duration, category, level, instructor expertise and teacher reputation. Revenue is additionally influenced by the number of transactions and the amount charged for each purchase. Therefore, a predictive framework should combine multiple dimensions rather than rely on descriptive statistics alone.

2.2 Research Problem

The central business problem addressed in this research is the absence of a unified analytical approach for predicting course demand and understanding revenue performance at EduPro. Management needs to identify which courses and categories attract learners, which factors are associated with enrollment, and where revenue opportunities exist. A second methodological problem is ensuring that predictive models do not accidentally use information that would only be known after the outcome occurs.

2.3 Research Questions

- Which course and teacher characteristics are associated with enrollment demand?
- How does revenue vary across course categories and course types?
- Which machine-learning algorithms perform best under cross-validation?
- How does feature engineering affect predictive performance?
- How can target leakage be identified and reduced in an educational analytics model?
- How can model results be converted into practical business recommendations?

2.4 Research Objectives

- Assess the quality and structure of EduPro datasets.
- Create an integrated master analytical dataset.
- Perform exploratory analysis of demand, revenue, teachers, transactions and users.
- Engineer business-relevant features for predictive modeling.
- Compare multiple regression algorithms using consistent validation.
- Investigate unexpectedly high model performance for leakage.
- Develop and tune a more defensible Random Forest enrollment model.
- Translate analytical findings into practical recommendations for EduPro.

3. Literature and Conceptual Foundation

3.1 Predictive Analytics in Education

Predictive analytics in education applies statistical and machine-learning techniques to historical information in order to estimate future or unknown outcomes. Common applications include student retention, course recommendation, learner success prediction, demand estimation and resource planning. In an online learning environment, predictive models can operate at several levels: learner, course, instructor, category and platform.

The present research focuses on course-level demand and revenue. This perspective is useful for management because course-level predictions can influence catalog design, promotional campaigns, instructor allocation and content investment. However, course-level outcomes are also affected by platform-wide seasonality and learner behavior. A robust system should therefore treat the current model as a foundation rather than a final production forecasting system.

3.2 Regression and Machine Learning

Regression methods estimate a numeric outcome from a set of predictors. Linear Regression provides an interpretable baseline and assumes a linear relationship between predictors and the target. Ridge Regression adds L2 regularization to reduce sensitivity to correlated predictors, while Lasso uses L1 regularization and can shrink coefficients toward zero. Tree-based ensemble models such as Random Forest and Gradient Boosting can represent nonlinear relationships and interactions without requiring explicit polynomial transformations.

Random Forest combines many decision trees and averages their predictions. Its ensemble structure can capture nonlinear patterns and is generally robust to different feature scales. Gradient Boosting builds trees sequentially, with later trees attempting to reduce errors made by earlier trees. These methods are appropriate candidates for the EduPro problem because course demand can depend on nonlinear combinations of price, rating, duration and instructor characteristics.

3.3 Model Evaluation

The study uses Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 . MAE expresses the average absolute prediction error in the target's units and is straightforward for managers to interpret. RMSE gives greater weight to larger errors and can reveal models that occasionally make substantial mistakes. R^2 measures the proportion of variance explained relative to a baseline model. No single metric is sufficient; therefore, all three should be considered together.

3.4 Cross-Validation

Five-fold cross-validation partitions the data into five subsets, trains on four and evaluates on the remaining subset, repeating the process so that each subset serves as validation data. Averaging across folds provides a more stable estimate than relying on one train-test split. For real deployment, time-based validation would be preferable if the objective is future forecasting, because random cross-validation can mix earlier and later observations.

4. Dataset Description and Data Architecture

4.1 Source Datasets

Dataset	Rows	Columns	Role
Courses	60	8	Course catalog and course attributes
Teachers	60	7	Instructor profile and quality variables
Transactions	10,000	7	Purchase/enrollment transactions and amounts
Users	3,000	5	Learner demographic information

The Courses dataset contains CourseID, CourseName, CourseCategory, CourseType, CourseLevel, CoursePrice, CourseDuration and CourseRating. Teachers contains TeacherID, TeacherName, Age, Gender, Expertise, YearsOfExperience and TeacherRating. Transactions records transaction identifiers, users, courses, dates, payment methods, amounts and teacher IDs. Users contains user identifiers, names, age, gender and email fields.

4.2 Data Relationships

CourseID provides the principal connection between the course catalog and transaction table. TeacherID links transaction records to instructors, while UserID connects transactions to learners. These relationships make it possible to move from transaction-level observations to course-level and category-level metrics. The analysis aggregates transaction counts to create EnrollmentCount and sums Amount to create CourseRevenue.

4.3 Data Quality

Dataset	Missing Values	Duplicates	Assessment
Courses	0	0	Clean
Teachers	0	0	Clean
Transactions	0	0	Clean
Users	0	0	Clean

All four datasets were reported as having zero missing values and zero duplicates. This means no imputation was required during the initial preparation stage. Nevertheless, clean structure does not automatically guarantee modeling validity. In particular, the temporal availability of features and whether a variable is derived from the target must be assessed separately.

4.4 Master Dataset Construction

Course-level transaction metrics were generated using groupby operations. EnrollmentCount was calculated as the number of transactions associated with each CourseID, while CourseRevenue was calculated as the sum of Amount. These metrics were merged into the course catalog. CategoryRevenue was then calculated by grouping course revenue by CourseCategory. The resulting course-level master dataset contained 60 rows and 11 columns before later transaction-level feature engineering.

5. Exploratory Data Analysis: Course Demand

5.1 Enrollment Distribution

EnrollmentCount ranged from 140 to 196 learners per course, with an average of approximately 166.67. The relatively narrow range indicates that course demand was broadly stable across the available catalog. This limited target variance is important for model interpretation: even a model with a moderate R^2 can produce useful absolute errors, while an apparently high R^2 can become suspicious if features directly encode the target.

5.2 Course Popularity

The analysis ranked courses by enrollment and identified a top course with approximately 196 enrollments. The distribution of enrollments suggests that the platform does not contain extreme demand concentration in a small number of courses. Instead, demand is relatively distributed. This may reflect the synthetic or balanced structure of the dataset and should be considered when generalizing the results to a real platform.

5.3 Category-Level Demand

Data Science was identified as a leading enrollment category, with approximately 900 total enrollments in the reported analysis. Category totals are influenced by the number of courses in each category, so category-level enrollment should ideally be interpreted alongside average enrollment per course. Management should avoid concluding that a category is intrinsically stronger solely because it has more courses.

5.4 Course-Level Demand by Difficulty

The reported level split was approximately 35.7% Beginner, 34.8% Advanced and 29.5% Intermediate. Beginner courses had a slight lead, but the overall distribution was balanced. This suggests that EduPro serves learners with different experience levels rather than being concentrated entirely on entry-level or advanced content.

5.5 Business Interpretation

- Stable enrollment across courses supports systematic demand monitoring rather than only focusing on outliers.
- Data Science's strong aggregate demand indicates a potential area for catalog expansion and targeted promotion.
- Balanced level participation supports maintaining a diversified learning portfolio.
- Course-level demand should be compared with price, rating and category size before investment decisions.
- Future studies should incorporate time-based enrollment data to distinguish sustained demand from short-term variation.

6. Exploratory Data Analysis: Revenue

6.1 Overall Revenue

Total course revenue was reported as approximately █911,323.47. Average revenue per course was approximately █15,188.72. The highest course revenue was approximately █85,416.60, while the lowest was █0. The presence of zero-revenue courses demonstrates why demand and revenue should be analyzed separately: a course can have learner activity without generating direct paid revenue if it is free.

6.2 Revenue by Category

Category	Approx. Revenue	Interpretation
Artificial Intelligence	█202,000	Highest reported category revenue
Business	█181,000	Second-highest contribution
Project Management	█169,000	Strong commercial performance
Programming	█77,000	Moderate contribution
Data Science	█76,000	Moderate contribution
Cybersecurity	█59,000	Moderate contribution
Digital Marketing	█56,000	Moderate contribution
Design	█43,000	Lower contribution
Finance	█28,000	Lower contribution
Web Development	█16,000	Lowest positive revenue
Machine Learning	█0	No reported revenue
Marketing	█0	No reported revenue

Artificial Intelligence generated the highest reported revenue at approximately █202,000, followed by Business and Project Management. These categories can be treated as priority segments for commercial analysis. However, revenue is partly driven by course price, so high revenue does not automatically imply high learner demand.

6.3 Free versus Paid Courses

The free-versus-paid analysis distinguishes catalog reach from direct monetization. Free courses can generate engagement and serve as acquisition channels, while paid courses contribute directly to transaction revenue. A useful management strategy is therefore to evaluate free courses using downstream conversion, repeat purchase and learner progression metrics rather than revenue alone.

6.4 Revenue Concentration

The top revenue-generating courses should be reviewed for pricing, ratings, instructor characteristics and category fit. High-revenue courses can become anchor products, but excessive concentration creates risk if demand changes. Portfolio analysis should therefore examine both revenue contribution and the stability of revenue across months.

7. Exploratory Data Analysis: Correlation and Relationships

7.1 Price and Enrollment

The reported correlation between CoursePrice and EnrollmentCount was approximately -0.16, indicating a weak negative relationship. This suggests that higher prices may be associated with somewhat lower enrollment, but price alone is not a strong explanatory factor. Management should therefore avoid assuming that price increases will necessarily reduce demand substantially across the catalog.

7.2 Rating and Enrollment

CourseRating showed a positive relationship with EnrollmentCount of approximately +0.29. Among the examined variables, this was the most meaningful reported relationship for demand. Higher ratings appear to be associated with stronger enrollment, supporting the importance of course quality, learner satisfaction and social proof.

7.3 Duration and Enrollment

CourseDuration showed a weak relationship with EnrollmentCount. This implies that course length alone does not strongly determine learner participation in the available data. Learners may value outcomes, quality, category relevance and perceived usefulness more than raw duration.

7.4 Correlation Summary

Relationship	Approx. Result	Business Meaning
CoursePrice ↔ Enrollment	-0.16	Weak negative association
CourseRating ↔ Enrollment	+0.29	Positive association; quality may support demand
CourseDuration ↔ Enrollment	Very weak	Duration alone has limited explanatory power
CourseRating ↔ CourseDuration	+0.21	Weak positive association

7.5 Caution About Correlation

Correlation measures association, not causation. A positive rating-enrollment relationship does not prove that increasing ratings will cause enrollment to increase. Ratings may themselves be influenced by course quality, learner selection, instructor quality or existing popularity. Controlled experiments and longitudinal data would be required to make stronger causal claims.

7.6 Implications for Feature Selection

The correlation findings support retaining CourseRating, CoursePrice and CourseDuration as candidate predictors, but the machine-learning stage should evaluate them jointly with category, level and teacher variables. Nonlinear models may capture patterns that pairwise correlation does not reveal.

8. Teacher, Transaction and User Analysis

8.1 Teacher Analysis

Teacher age ranged from approximately 27 to 50 years, while YearsOfExperience ranged from 1 to 24 years. TeacherRating ranged from approximately 1.05 to 4.97. The experience-versus-rating plot was used to examine whether more experienced teachers consistently receive higher ratings. Such an association should not be assumed automatically; instructor quality can depend on communication, course design and subject expertise.

8.2 Teacher Features

YearsOfExperience and TeacherRating were incorporated into the machine-learning feature set. These variables provide a bridge between course characteristics and instructor quality. An instructor with extensive experience may provide stronger course structure, but experience is not a direct guarantee of learner demand.

8.3 Transaction Trend

Monthly transaction revenue fluctuated during the year. The reported highest month was November at approximately ■83,500, while February was approximately ■67,000. Overall monthly revenue remained within a relatively stable range of about ■67,000 to ■84,000. The observed pattern indicates periodic variation rather than a prolonged decline.

8.4 Business Interpretation of Seasonality

If the same seasonal pattern is observed in real operational data, marketing teams could use lower-performing periods for targeted promotions and use high-performing periods for premium campaigns. However, a single year of transactions is insufficient to establish reliable seasonality. Multiple years of observations are preferable.

8.5 User Demographics

The reported user age distribution was approximately 15 to 35 years, suggesting that EduPro primarily serves teenagers and young adults in the supplied dataset. Gender participation was described as nearly balanced, with female users slightly outnumbering male users.

The analysis initially displayed UserID frequency. Because UserID is unique by definition, this chart does not provide meaningful behavioral insight. Future user analysis should instead examine age bands, repeat transactions, preferred categories, average spending, purchase frequency and conversion behavior.

8.6 Privacy and Responsible Analytics

User email and identifying fields are not necessary for course-demand modeling and should be excluded from analytical features. A production system should minimize personally identifiable information, apply access controls and use aggregated or anonymized learner attributes wherever possible.

9. Data Preparation and Master Data Construction

9.1 Transaction-Level Merge

The project also constructed a transaction-level master dataset by merging Courses with Transactions using CourseID and then joining Teachers using TeacherID. This produced a detailed analytical table containing course, transaction and teacher information. The resulting structure is useful for feature engineering but must be handled carefully because the same course-level target can repeat across many transaction rows.

9.2 Derived Outcomes

EnrollmentCount was generated by grouping transaction records by CourseID and counting UserID. CourseRevenue was generated by grouping the same records and summing Amount. CategoryRevenue was then generated by grouping CourseRevenue by CourseCategory. These transformations are analytically valid for descriptive reporting, but if the resulting metrics are used as predictors of themselves, target leakage occurs.

9.3 Data Cleaning Operations

- Column names were stripped of leading and trailing whitespace.
- Transaction dates were converted to datetime format.
- Missing and duplicate values were checked.
- Numeric and categorical fields were identified for modeling.
- Course-level and category-level metrics were created using grouped aggregation.
- Zero denominators were protected when calculating revenue per enrollment.

9.4 Analytical Grain

The analytical grain is critical. The original transaction table has one row per transaction, while the initial master course table has one row per course. The feature-engineered dataset has 10,000 transaction-level rows, but several course-level outcomes are repeated on every transaction belonging to the same course. This repeated-target structure can make a random train-test split appear more accurate than a genuinely future-facing forecast.

9.5 Recommended Production Grain

For future deployment, the preferred grain should be defined before model training. If the target is next-month course enrollment, each training row should represent a course-month, with predictors constructed only from information available before that month. This would prevent future transactions from being included in historical features and would align the model with the actual business forecasting task.

10. Feature Engineering

10.1 PriceBand

Course price was transformed into Free, Low, Medium and High bands. This allows the model to capture nonlinear price effects. A free course is structurally different from a paid course, and a high-priced course may have a different demand response than a low-priced course.

10.2 DurationBucket

Course duration was divided into Short, Medium, Long and Very Long groups. The feature provides a categorical representation of content length and can capture threshold effects that a single continuous duration variable may not represent.

10.3 RatingTier

Course ratings were converted into Low, Medium, High and Top tiers. This creates a business-friendly interpretation of course quality and can help tree-based models capture nonlinear relationships.

10.4 ExperienceBucket

Teacher experience was grouped into Junior, Mid, Senior and Expert. This reduces dependence on a precise numeric threshold and produces interpretable instructor segments.

10.5 TeacherRatingTier

Teacher ratings were transformed using the same tier logic. This feature provides an instructor-quality segment that can be compared with course ratings.

10.6 ExpertiseMatch

ExpertiseMatch was created by comparing teacher expertise with course category. A value of 1 indicates a text match and 0 indicates no match. This is a useful concept, but the exact string comparison is strict. In a production system, a subject taxonomy or mapping table would be more reliable than exact text equality.

10.7 RevenuePerEnrollment

RevenuePerEnrollment was calculated as CourseRevenue divided by EnrollmentCount, with zero denominators replaced to prevent division errors. This feature is highly informative descriptively, but it is problematic as a predictor of CourseRevenue because it contains the target itself. The project therefore removed it from the enrollment and revenue model feature matrices in the main training steps.

10.8 IsFree

IsFree is a binary indicator equal to 1 when CourseType is Free and 0 otherwise. It duplicates some information contained in CourseType but provides a direct numeric representation useful for certain models.

10.9 Feature Engineering Summary

Feature	Type	Purpose
PriceBand	Categorical	Price segmentation
DurationBucket	Categorical	Duration segmentation
RatingTier	Categorical	Course quality segmentation
ExperienceBucket	Categorical	Teacher experience segmentation
TeacherRatingTier	Categorical	Teacher quality segmentation
ExpertiseMatch	Binary	Subject alignment
RevenuePerEnrollment	Numeric	Unit economics; excluded when leakage risk exists
IsFree	Binary	Free-course indicator

11. Machine Learning Methodology

11.1 Target Variables

Three targets were defined: EnrollmentCount, CourseRevenue and CategoryRevenue. EnrollmentCount measures course demand, CourseRevenue measures direct course revenue, and CategoryRevenue aggregates revenue across courses within a category.

11.2 Feature Encoding

Categorical variables including CourseCategory, CourseType, CourseLevel, PriceBand, DurationBucket, RatingTier, ExperienceBucket and TeacherRatingTier were encoded using LabelEncoder. The resulting encoded fields were appended to the modeling table. For a production system, one-hot encoding or target-safe categorical methods may be preferable because label encoding can introduce artificial ordinal relationships into linear models.

11.3 Algorithms

Model	Purpose
Linear Regression	Interpretable linear baseline
Ridge Regression	Linear model with L2 regularization
Lasso Regression	Linear model with L1 regularization
Random Forest	Ensemble of randomized decision trees
Gradient Boosting	Sequential tree-based boosting

11.4 Scaling

StandardScaler was applied for linear regression, Ridge and Lasso because these methods can be sensitive to feature scale. Tree-based models were trained on the unscaled feature matrix because tree splits do not require standardized numeric ranges.

11.5 Five-Fold Cross-Validation

The initial helper function used shuffled five-fold cross-validation with random_state 42. MAE, RMSE and R² were calculated across folds. The best model was selected using the highest average R² and then fitted on the full feature matrix before being saved with joblib.

11.6 Model Storage

Models were saved under /content/models using files such as best_Enrollment_Count.pkl, best_Course_Revenue.pkl and best_Category_Revenue.pkl. A production implementation should also save preprocessing objects, feature names, model version, training date and data schema to prevent scoring inconsistencies.

12. Initial Model Results

12.1 Enrollment Count

Model	MAE	RMSE	R ²
Linear Regression	8.36	10.33	0.311
Ridge Regression	8.36	10.33	0.311
Lasso Regression	8.39	10.36	0.308
Random Forest	0.00	0.00	1.000
Gradient Boosting	1.34	1.71	0.981

Random Forest reported a perfect R² of 1.000, with zero MAE and RMSE. Gradient Boosting also produced an extremely high R² of approximately 0.981. Such results are not automatically evidence of an excellent production model; they require investigation of feature construction and data leakage.

12.2 Course Revenue

Model	MAE	RMSE	R ²
Linear Regression	1122.85	1915.35	0.994
Ridge Regression	1122.04	1915.36	0.994
Lasso Regression	1122.74	1915.35	0.994
Random Forest	0.00	0.00	1.000
Gradient Boosting	2.47	4.57	1.000

Course revenue was also predicted almost perfectly by tree models. The strong performance is expected to some extent because CoursePrice is directly connected to transaction Amount in the supplied dataset. Nevertheless, the near-perfect results should not be interpreted as proof of future forecasting accuracy.

12.3 Category Revenue

Model	MAE	RMSE	R ²
Linear Regression	6,164,378.81	7,794,804.65	0.516
Ridge Regression	6,164,177.54	7,794,798.81	0.516
Lasso Regression	6,164,378.75	7,794,804.65	0.516
Random Forest	0.00	0.00	1.000
Gradient Boosting	131,170.19	176,568.26	1.000

The category revenue results reinforce the leakage concern. CategoryRevenue is an aggregate of CourseRevenue and is repeated across transaction rows. A model can therefore identify the category-level outcome through repeated or closely related information. This is why model evaluation must be tied to the intended prediction scenario.

13. Model Performance Comparison and Feature Importance

13.1 Comparative Interpretation

The initial model comparison demonstrates the flexibility of tree-based ensembles. Linear, Ridge and Lasso regression achieved substantially lower R^2 for enrollment than the tree-based models. For course revenue, even the linear models achieved very high R^2 because price and transaction amount are strongly connected in the supplied data. Category revenue showed a moderate linear R^2 but perfect tree-model results, again suggesting that repeated aggregate structure may be enabling the model to reconstruct the target.

13.2 Reported Feature Importance

Target	Reported leading features	Interpretation
Enrollment Count	CourseDuration, RatingTier, CourseRating, CourseCategory	Demand is associated with duration, quality and category structure in the supplied importance analysis.
Course Revenue	CoursePrice	Price dominates revenue prediction, consistent with the transaction structure.
Category Revenue	RevenuePerEnrollment, CoursePrice	Revenue-derived information dominates; this should not be used for unbiased forecasting.

Feature importance is model-specific and should not be interpreted as causal influence. In addition, importance values can become misleading when features are derived from the target or are strongly correlated with it. The reported importance chart is therefore most useful as a diagnostic tool for understanding the initial model, not as a final causal ranking.

13.3 Recommended Importance Methods

Future analysis should compare impurity-based importance with permutation importance and SHAP values. Permutation importance evaluates the effect of disrupting each feature on validation performance. SHAP provides local and global contribution estimates and can show whether higher or lower values push predictions upward or downward.

13.4 Business Value of Interpretability

Management needs to understand not only what the model predicts but why. A transparent feature analysis can help identify whether course quality, price, duration, teacher characteristics or category are associated with demand. Interpretability also supports model governance and helps analysts identify suspicious predictors before deployment.

14. Target Leakage: Diagnosis and Correction

14.1 What Is Target Leakage?

Target leakage occurs when a model receives information that would not be legitimately available at the moment the prediction is supposed to be made. Leakage can occur through direct target variables, post-outcome aggregates, future transactions, identifiers, or derived measures that encode the target.

14.2 Leakage in the EduPro Dataset

The clearest example is RevenuePerEnrollment. It is calculated as CourseRevenue divided by EnrollmentCount. When CourseRevenue is the target, this feature mathematically contains the target and should not be used for revenue prediction. Similarly, when EnrollmentCount is the target, a feature based on post-period transaction counts can encode the outcome.

The transaction-level modeling table also repeats course-level EnrollmentCount and CourseRevenue across many rows. Randomly splitting these repeated rows can place records from the same course into both training and validation sets. The model may therefore learn course-specific patterns rather than generalize to genuinely unseen courses or future periods.

14.3 Why Perfect R^2 Is Suspicious

A perfect R^2 of 1.000 can occur legitimately in simple deterministic problems, but it is unusual for behavioral demand prediction. The zero MAE and zero RMSE reported for Random Forest strongly motivated a leakage review. In practical analytics, a model that predicts real-world learner behavior with no error should be treated as a diagnostic signal until independently validated.

14.4 Leakage Prevention Rules

- Define the prediction date before constructing features.
- Use only data available before that date.
- Do not use target-derived ratios as predictors of the same target.
- Aggregate historical behavior only up to the cutoff date.
- Group or time-split data when repeated entities exist.
- Keep validation data completely isolated from preprocessing and feature construction.
- Compare random cross-validation with time-based validation for forecasting tasks.

14.5 Research Significance

The leakage investigation is an important part of the research contribution. Rather than presenting the highest numerical score as the final result, the study demonstrates why predictive analytics requires methodological scrutiny. The improved enrollment model provides a more realistic benchmark.

15. Improved Enrollment Model and Hyperparameter Tuning

15.1 Rationale

Because enrollment values ranged only from 140 to 196 and the initial model showed suspiciously perfect performance, an improved model was developed using a restricted predictor set. The selected variables were CoursePrice, CourseDuration, CourseRating, YearsOfExperience, TeacherRating, CourseCategory_enc, CourseType_enc and CourseLevel_enc.

15.2 Hyperparameter Search

Parameter	Values Tested
n_estimators	50, 100, 200
max_depth	3, 5, None
min_samples_leaf	2, 4, 6
Cross-validation	5 folds
Scoring	R ²

GridSearchCV evaluated combinations of tree count, tree depth and minimum leaf size. The selected configuration was n_estimators = 100, max_depth = 3 and min_samples_leaf = 2. Limiting tree depth and requiring multiple observations per leaf reduces the ability of the model to memorize individual training observations.

15.3 Final Performance

Metric	Result
Algorithm	Random Forest Regressor
Target	EnrollmentCount
Training R ²	0.442
5-Fold CV R ²	0.453
Training MAE	7.50
5-Fold CV MAE	7.51
Training RMSE	9.30
Best parameters	max_depth=3; min_samples_leaf=2; n_estimators=100

The training R² of 0.442 and cross-validation R² of 0.453 are much lower than the earlier perfect score, but this is not a failure. The revised result is more credible because the model is evaluated with a feature set that is less directly tied to the target. The similarity between training MAE and cross-validation MAE is also reassuring because it indicates limited overfitting under the selected validation scheme.

15.4 Interpretation

The model explains approximately 45% of the variance under five-fold cross-validation. The remaining unexplained variance indicates that course and teacher attributes alone are insufficient to fully predict demand. Variables such as marketing exposure, learner intent, course launch date, reviews, search impressions, recommendation placement and historical demand trends would likely improve forecasting.

16. Discussion of Results

16.1 Demand Drivers

The exploratory analysis and model development jointly indicate that demand is multidimensional. Course rating had a positive relationship with enrollment, while price showed a weak negative relationship. The improved model additionally includes duration, category, level and teacher characteristics. These variables capture product positioning and perceived quality but cannot fully represent promotional or behavioral effects.

16.2 Revenue Drivers

Course revenue is strongly related to price in the supplied transaction structure. This is intuitive because the transaction Amount is closely tied to CoursePrice. However, revenue forecasting in a production environment should predict future transaction volume and expected price separately or use a future-period revenue target. Otherwise, the model may simply reconstruct accounting totals that are already known.

16.3 Model Selection

Random Forest was the strongest initial performer, but model selection should not be based only on R^2 . The improved Random Forest was selected because it balanced nonlinear modeling capacity with constraints that reduce memorization. Gradient Boosting can also be useful, especially after careful tuning, but the selected model is easier to explain and provides a practical baseline.

16.4 Validation Quality

The research highlights an important distinction between random cross-validation and forecasting validation. Random folds are useful for estimating generalization among similarly distributed records, but future forecasting requires temporal separation. A better production design would train on earlier months and validate on later months, with rolling or expanding windows.

16.5 Statistical and Practical Significance

An R^2 of 0.453 does not mean the model is useless. With a narrow target range, a prediction error of about 7.5 enrollments may still be operationally useful if managers need broad demand categories rather than exact counts. Conversely, a model with R^2 of 1.000 may be unusable if its inputs are unavailable at prediction time. Therefore, practical utility depends on accuracy, timing, interpretability and data availability.

16.6 Research Contribution

The key contribution is an end-to-end framework that combines descriptive analytics with leakage-aware predictive modeling. The study shows how raw platform data can be transformed into managerial insights and how suspiciously high model scores should trigger validation rather than immediate acceptance.

17. Business Recommendations and Decision Framework

17.1 Course Portfolio Management

- Prioritize categories with sustained enrollment and strong revenue rather than relying on one metric.
- Review high-revenue courses for pricing, quality and instructor characteristics to understand what makes them successful.
- Use low-revenue categories as candidates for repositioning, promotion, redesign or portfolio rationalization.
- Maintain a balanced course portfolio across Beginner, Intermediate and Advanced levels.

17.2 Pricing Strategy

The weak negative association between price and enrollment suggests that price should be tested carefully. EduPro can experiment with discounts, bundles, introductory pricing and premium positioning. A/B testing would be more appropriate than interpreting correlation as a causal price effect.

17.3 Course Quality

The positive association between rating and enrollment suggests that quality improvement may support demand. Management should monitor learner feedback, completion rates, review sentiment and instructor performance. Courses with low ratings but strategically important topics may deserve redesign rather than removal.

17.4 Marketing Allocation

The monthly transaction trend suggests periods of lower and higher activity. Marketing resources can be shifted toward lower-performing periods, while high-performing periods can be used for cross-selling and premium campaigns. The strategy should be validated using multiple years of data.

17.5 Instructor Strategy

Teacher experience and rating can be used as supporting indicators when evaluating course performance. However, management should avoid assuming that experience alone drives demand. Instructor-course expertise alignment, learner reviews and completion outcomes should be incorporated.

17.6 Predictive Dashboard

A practical dashboard should provide course demand forecasts, revenue forecasts, category comparisons, model confidence or error ranges, feature importance, and alerts for unusual changes. The dashboard should distinguish historical actuals from predictions and clearly display the forecast horizon.

17.7 Decision Matrix

Situation	Recommended Action
High demand + high revenue	Protect, promote and consider advanced versions
High demand + low revenue	Review pricing, free/paid strategy and conversion
Low demand + high revenue	Protect profitability; investigate customer concentration
Low demand + low revenue	Consider redesign, targeted promotion or retirement
High rating + low demand	Investigate discoverability and marketing
Low rating + low demand	Prioritize quality improvement before promotion

18. Limitations, Future Research and Implementation Roadmap

18.1 Limitations

- The dataset contains only one year of transaction dates, limiting robust seasonality analysis.
- The target enrollment range is relatively narrow, which constrains achievable variance explanation.
- The transaction-level modeling structure repeats course-level outcomes and can create leakage risks.
- Label encoding may introduce artificial ordering for categorical variables in linear models.
- RevenuePerEnrollment is target-derived and should not be used for direct revenue prediction.
- User-level behavioral features are not extensively incorporated into the final enrollment model.
- No external marketing, search, campaign, review or platform-placement variables are available.
- The current validation is randomized; true future forecasting requires temporal validation.

18.2 Future Feature Development

A stronger forecasting dataset should be constructed at course-month level. Candidate predictors include lagged enrollments, rolling seven- and thirty-day demand, historical conversion rate, page views, impressions, search ranking, recommendation exposure, discounts, campaign type, review volume, average review score, completion rate, refund rate and repeat learner rate.

18.3 Advanced Modeling

Future experiments can compare Random Forest with XGBoost, LightGBM, CatBoost and regularized generalized additive models. Hyperparameter tuning should be performed inside cross-validation. Feature importance should be complemented by permutation importance and SHAP. Prediction intervals should be considered so managers can understand uncertainty rather than relying on a single point forecast.

18.4 Deployment Roadmap

Phase	Activity	Output
1	Data pipeline	Validated course-month analytical table
2	Feature store	Leakage-safe historical predictors
3	Model training	Candidate models and cross-validation results
4	Temporal validation	Future-period performance benchmark
5	Dashboard	Forecasts, KPIs and alerts
6	Monitoring	Drift, error and data-quality monitoring
7	Retraining	Scheduled model updates

18.5 Governance

A production model should record data version, feature definitions, model version, training period, validation results and prediction date. Automated checks should detect missing fields, unexpected categories, distribution shifts and leakage-prone columns. Business users should be able to distinguish predictions from actual results.

19. Conclusion, References and Appendix

19.1 Conclusion

This research developed a predictive analytics framework for EduPro, combining course, teacher, transaction and user data to investigate enrollment demand and revenue behavior. The data were clean, with zero missing values and duplicates reported across the four source datasets. Exploratory analysis identified stable course-level enrollment, strong revenue contributions from Artificial Intelligence, Business and Project Management, a positive association between course rating and enrollment, and moderate monthly revenue variation.

The machine-learning stage compared Linear Regression, Ridge, Lasso, Random Forest and Gradient Boosting. Initial models produced exceptionally high performance, including perfect R^2 values for several targets. Rather than treating these scores as final evidence of predictive quality, the study investigated target leakage and repeated aggregate information. This methodological step is central to the reliability of the research.

The improved enrollment model used a restricted set of course and teacher predictors and a tuned Random Forest Regressor. The final configuration used 100 estimators, maximum depth of 3 and minimum leaf size of 2. It achieved a training R^2 of 0.442 and five-fold cross-validation R^2 of 0.453, with cross-validation MAE of approximately 7.51 enrollments. The result is more credible as a baseline because it reduces the ability to memorize target-derived information.

Overall, EduPro can use predictive analytics to support course planning, quality improvement, pricing experiments, marketing allocation and portfolio decisions. For a production forecasting system, the next priority should be course-month temporal modeling with lagged historical features and future-period validation. The research demonstrates that trustworthy predictive analytics is not simply about maximizing a score; it requires sound data construction, leakage control, appropriate validation and alignment with the business decision being supported.

19.2 References

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- Project dataset: EduPro Online Platform workbook containing Courses, Teachers, Transactions and Users sheets.

19.3 Appendix: Key Project Artifacts

Artifact	Description
Master dataset	Course-level analytical dataset with enrollment and revenue metrics
Feature-engineered dataset	Transaction-level dataset with engineered categorical and numeric features
Enrollment model	best_Enrollment_Count.pkl
Revenue model	best_Course_Revenue.pkl
Category revenue model	best_Category_Revenue.pkl
Final ML dataset	master_ml_final.xlsx
Model evaluation	MAE, RMSE and R^2 using five-fold cross-validation
Improved model	Tuned Random Forest enrollment model

End of Research Paper